BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION SP/2024)

CLASS: BTech/IMSC **SEMESTER : II**
BRANCH: All/PHYSICS **SESSION : SP/2024**

SUBJECT: MA107 MATHEMATICS – II

TIME: 02 Hours **FULL MARKS: 25**

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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		Marks	CO	BL
Q.1(a)	Find the Wronskian of xe^x and x . Check whether they are linearly independent or not.	[2]	1	1
Q.1(b)	Solve the following differential equation: $$y^{iv} - 3y'' - 4y = 0.$$	[3]	1	2
Q.2(a)	Find the particular solution of : $$(\mathrm{D}^2 - 1)\mathrm{y} = \cos 2\mathrm{x} + \mathrm{a}^{\mathrm{x}}$$	[2]	1	1
Q.2(b)	Solve the following differential equation: $$[D-1]^2 y(x) = e^x sin2x$$	[3]	1	2
Q.3	Find the power series solution about $x = 0$ of the differential equation: $$y'' - xy' - y = 0$$	[5]	2	1
Q.4(a)	Show that when n is an integer, $J_{-n}(x) = (-1)^n J_n(x)$	[2]	2	2
Q.4(b)	Express x^3 in terms of Legendre polynomials and hence using orthogonality condition evaluate: $$\int_{-1}^{1} x^3 P_3(x)dx$$	[3]	2	1
Q.5	Find the Fourier series representation of a periodic function given by: $1 - x^2, -\pi < x < \pi.$	[5]	3	1

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